

LETHBRIDGE, AB, Canada

WMO: 712670

Lat: 49.626N

Lon: 112.794W

Elev: 929

StdP: 90.65

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.2	-25.5	-33.7	0.2	-28.9	-29.9	0.3	-25.4	18.5	4.6	17.0	4.6	3.0	230	0.740

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.6	31.7	16.6	29.6	16.2	27.7	15.8	18.8	27.1	17.7	26.3	16.7	25.4	5.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	12.7	22.4	14.7	11.7	21.1	13.6	10.8	19.8	57.4	27.1	53.7	26.2	50.5	25.5	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.8	13.8	12.2	DB	-34.7	35.0	3.7	2.1	-37.4	36.5	-39.5	37.8	-41.6	38.9	-44.2	40.5	(4)
(5)			WB	-34.9	21.3	3.6	1.6	-37.5	22.5	-39.6	23.4	-41.6	24.3	-44.2	25.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.7	-6.1	-5.2	-0.5	5.3	10.6	14.6	18.3	17.6	13.0	6.3	-0.7	-5.7	(6)	
	DBStd	10.90	9.46	8.86	8.08	5.06	4.39	3.24	3.06	3.42	4.60	5.50	7.75	8.75	(7)	
	HDD10.0	2425	499	427	328	153	46	3	1	1	23	136	322	486	(8)	
	HDD18.3	4698	757	660	584	392	242	117	38	54	166	373	570	744	(9)	
	CDD10.0	854	1	0	2	11	64	143	259	238	114	22	2	0	(10)	
	CDD18.3	86	0	0	0	0	2	7	38	33	7	0	0	0	(11)	
	CDH23.3	1835	0	0	1	6	74	148	696	676	228	7	0	0	(12)	
	CDH26.7	627	0	0	0	0	13	32	249	260	72	0	0	0	(13)	
(14) Wind	WSAvg	5.1	5.8	5.3	5.4	5.4	5.1	4.8	4.3	4.1	4.6	5.5	5.7	5.8	(14)	
(15) Precipitation	PrecAvg	425	24	16	28	40	54	67	45	44	46	18	18	23	(15)	
	PrecMax	766	59	37	78	169	134	169	123	143	140	46	51	58	(16)	
	PrecMin	243	5	2	5	0	10	3	7	4	4	0	0	4	(17)	
	PrecStd	111	13	8	16	33	30	42	32	32	35	13	14	14	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	13.8	18.2	24.0	28.4	30.3	34.3	34.5	32.0	24.4	18.0	12.2	(19)	
		MCWB	6.3	5.9	8.1	11.5	14.4	17.0	17.6	16.9	15.2	12.0	8.7	5.2	(20)	
	2%	DB	9.7	10.6	15.2	20.4	25.5	27.0	31.7	31.8	28.8	20.8	14.4	9.4	(21)	
		MCWB	4.2	4.3	7.0	9.7	13.4	15.9	17.5	16.1	14.3	10.9	7.2	3.8	(22)	
	5%	DB	7.6	8.2	12.8	17.4	22.7	24.8	29.5	29.6	26.0	17.9	11.8	7.4	(23)	
		MCWB	2.9	3.0	5.7	8.3	12.2	15.0	17.2	15.9	13.5	9.6	5.8	2.6	(24)	
	10%	DB	5.7	6.1	10.2	14.7	20.1	22.7	27.4	27.4	23.0	15.1	9.3	5.4	(25)	
		MCWB	1.9	1.6	4.2	6.9	10.9	13.9	16.9	15.4	12.6	7.9	4.1	1.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	6.6	9.0	12.0	15.5	19.1	21.0	19.1	16.5	13.2	9.9	6.1	(27)	
		MCDB	12.1	12.9	16.7	22.0	25.7	26.6	29.4	28.4	27.9	22.0	16.9	10.9	(28)	
	2%	WB	4.5	4.8	7.4	10.3	14.0	17.2	19.5	17.8	15.1	11.4	7.6	4.2	(29)	
		MCDB	9.1	10.1	14.4	19.5	23.7	24.5	27.4	27.5	26.0	19.7	13.7	8.8	(30)	
	5%	WB	3.1	3.2	6.0	8.7	12.7	15.9	18.5	16.9	14.1	9.9	5.9	2.8	(31)	
		MCDB	7.2	7.9	12.3	16.6	21.5	23.1	26.6	26.5	24.2	17.2	11.2	7.0	(32)	
	10%	WB	2.0	1.8	4.4	7.2	11.5	14.6	17.5	16.1	12.9	8.4	4.4	1.4	(33)	
		MCDB	5.5	5.9	9.8	14.0	19.0	21.3	25.7	25.4	22.3	14.6	9.1	5.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	11.6	12.1	13.1	13.4	12.9	15.6	16.2	15.0	13.0	11.3	10.8	(35)	
		MCDBR	11.5	13.3	15.8	18.5	18.3	17.2	19.8	21.0	21.0	17.2	13.9	11.3	(36)	
	5% WB	MCWBR	7.7	8.6	9.4	10.0	8.9	8.2	8.7	8.6	9.3	9.1	8.6	7.6	(37)	
		MCDWR	11.4	13.5	15.6	17.3	16.8	15.5	17.2	18.1	19.1	16.4	13.7	11.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.250	0.259	0.284	0.320	0.333	0.330	0.333	0.350	0.312	0.280	0.257	0.234	(40)	
		taud	2.454	2.486	2.481	2.388	2.406	2.460	2.465	2.393	2.471	2.540	2.503	2.456	(41)	
	Ebn at Noon		832	912	938	928	925	928	919	885	891	872	818	806	(42)	
		Edh at Noon	59	73	88	108	111	106	104	107	89	69	56	50	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.24	2.24	3.50	4.84	5.77	6.42	7.07	5.61	3.99	2.48	1.39	0.98	(44)	
	RadStd		0.10	0.17	0.29	0.33	0.48	0.47	0.37	0.34	0.33	0.25	0.11	0.10	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (66 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page